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Fusing Delta Learning and Machine Learned Interatomic Potentials for Efficient High Throughput Calculation of Surface Kinetic Parameters

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Abstract

Heterogeneous catalysis is critical in most industrial chemical processes. Microkinetic models can be used to greatly facilitate optimization of catalyst design and process conditions, but require thermochemical and kinetic parameters for all relevant species and reactions. Our software Pynta enables fully automated calculation of thermochemical and kinetic parameters, however, the computational cost of density functional theory (DFT) calculations makes it difficult to calculate kinetic parameters at scale. In this work, we combine finetuning of the MACE multi-head v0 model with graph-based delta learning of stationary points using subgraph isomorphic decision trees (SIDT). We first generate a target chemical space on Pt111 by using the Reaction Mechanism Generator (RMG) software to generate all possible surface reactions between a set of 153 chemically adsorbed species containing C,H,N, and O. Applying Pynta to this reaction set we are able to generate geometries for many gas phase species, adsorbates and transition states in this chemical space allowing us to efficiently sample near stationary points for finetuning and stationary points for delta learning. In particular, we adapt Pynta's harmonically force saddle point search (HFSP) algorithm to enable efficient and reliable sampling of near transition state points. With this training data we show that SIDT-driven delta learning of the foundation MLIP alone can achieve similar accuracies on enthalpies to foundation model finetuning approaches. However, we show that by combining the two approaches into one framework we are able to significantly improve our accuracies over either

approach allowing us to achieve 0.07 eV MAE relative to DFT on test barrier heights. Applying this framework across all reactions in our chemical space, we are able to obtain near DFT accuracy rate coefficients for 3111 reactions on Pt(111).

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Introduction

Heterogeneous catalysis is a critical aspect of most industrial chemical processes. Optimizing these chemical processes as a function of the catalyst and process conditions is of high interest. Microkinetic models can greatly facilitate optimizing process conditions for a given catalyst. However, catalytic reaction mechanisms are complex involving many reactions and species and kinetic parameters vary significantly between catalysts.

Tools like the Reaction Mechanism Generator software (RMG)¹⁻³ can be used to auto-

matically explore the chemical space and generate microkinetic models. These models can then be used by kinetics simulation software like ReactionMechanismSimulator.jl⁴ and Cantera⁵ to optimize reaction processes. However, microkinetic mechanisms are only as good as their kinetic parameters. While, these parameters can be calculated with complex sequences of density function theory (DFT) calculations, running many such calculations can be very computationally expensive. If RMG does not have an exact value in its database for kinetic parameters it estimates them using property prediction models such as: heuristic methods,⁶⁻⁹ subgraph isomorphic decision tree (SIDT) driven approaches,¹⁰⁻¹² and deep neural network (DNN) driven approaches.^{13,14} However, due to limited training data, these estimators have limited accuracy, which is especially important for sensitive/critically important reactions in these mechanisms. In order to improve these estimators we need large quantities of accurate kinetic parameters for training.

To meet these challenges, we need to be able to rapidly and readily calculate many kinetic parameters from ab initio methods. Our software Pynta^{15,16} enables efficient, robust, and automated calculation of kinetic parameters on surfaces. However, even when done efficiently, quantum chemistry calculations on surfaces are computationally expensive making it challenging to calculate kinetic parameters for even tens of reactions let alone hundreds or thousands.

Machine learned interatomic potentials (MLIPs)¹⁷⁻¹⁹ trained on sets of density functional theory (DFT) calculation provide a much faster alternative than pure DFT based approaches. Schaaf et al. developed automatic MLIP sampling and training protocols that enabled efficient resolution of the reaction network for hydrogenation of carbon dioxide to methanol on indium oxide. More recently, finetuning pre-trained foundation models has been shown to reduce the amount of new DFT data needed to reach a given target accuracy, compared with from scratch training^{21,22} However, many foundation models show degraded performance for higher-energy and reactive configurations on surfaces, plausibly reflecting limited representation of such regions in available training data and additional surface-specific complexities.²³

Price et al. demonstrated that Pynta can be used to enable highly effective sampling of both wells and transition states allowing for iterative refinement of MLIPs to accuracy on reaction barriers for 53 reactions on Ag(111). In particular, it showed that Pynta’s harmonically forced saddle point search (HFSP) method developed for finding transition states can be used provide a convenient, reliable, and effective way to sample near saddle points, which is traditionally very challenging. However, the iterative framework ran into major difficulties with model degradation and catastrophic forgetting, which motivated the training of reaction class separated MLIPs.²²

When calculating kinetic parameters using statistical mechanics and transition state theory, however, we are primarily interested in the energies of stationary points (wells and transition states) that can be represented as molecular graphs. In any system the chemical space of all stationary points is dramatically smaller in dimension than the space of all possible molecular geometries. So provided an MLIP has decent performance and can find the stationary points of interest (i.e., the PES is qualitatively correct), it should require dramatically fewer training points mapping from the MLIP stationary point energies to DFT stationary point energies than a finetuning procedure targeting the entire geometrical domain of the MLIP. While this kind of mapping is now commonly referred to as delta learning, similar frameworks significantly predate widespread use of machine learning in chemistry. Bond additivity correction (BAC) schemes^{24–29} are still commonly and effectively used to correct gas-phase calculated enthalpies of formation to experimental values by summing corrections associated with the bond type (e.g. N=C, O-H) of each bond. BACs are constructed by fitting these corrections to the difference between a carefully curated database of experimental values and values calculated using the chosen quantum chemistry method.²⁸ While BACs have achieved the most widespread adoption in the field, other more traditional delta learning approaches also exist in literature. Ramakrishnan et al. leveraged kernel ridge regression delta learning to approach G4MP2 accuracy from DFT calculations. Xu et al. and Yang et al. combined aspects of CBH schemes, BACs, and traditional machine learning

techniques to correct from DFT to CCSD(T).

While BACs have been demonstrated to be incredibly effective, they are limited by their rigid fitting structure. Subgraph isomorphic decision trees (SIDTs) are a machine learning approach that predicts molecular properties by descending molecular substructures down a tree of increasing specific substructures and making predictions based on which tree nodes are matched.^{10,11} SIDTs have inherent error cancellation on relative properties, straightforward and robust uncertainty estimation, and because they are trees of molecular substructures they are inherently interpretable and have good uncertainties.^{10-12,16,33,34} Our software, PySIDT was recently demonstrated to outperform both state-of-the-art deep neural network methods like Chemprop^{13,14} and state-of-the-art sparse methods like XGBoost³⁵ in vanilla chemical property prediction learning comparisons in both sparse and dense data regimes.¹¹ Additionally, SIDTs can be structured identically to BACs, except that they can be more or less specific as datasets allow (rather than being a fixed fitting framework).

In this work we will develop a combined finetuning and SIDT-driven delta learning framework for rapidly and accurately predicting rate coefficients on Pt(111). First we will leverage RMG to generate a broad chemical space of CHON stationary points including 153 possible adsorbates and 8451 possible reactions. We will leverage the MACE multi-head v0 MLIP and Pynta to easily generate stationary point and near stationary point training data for both finetuning the MLIP and training a SIDT delta model. After a single sampling and finetuning/training iteration we will show that combining the finetuned MLIP with the SIDT delta model we can predict stationary point energies and barriers within ~ 0.07 eV MAE relative to DFT. Lastly, running Pynta with the finetuned model we are able to find transition states for 3111 reactions allowing this framework to provide us near-DFT accuracy rate coefficients for 3111 reactions on Pt(111).

Methods

Density Functional Theory

We ran all DFT calculations on a $3 \times 3 \times 4$ periodic Pt(111) slab using BEEF-vdW with PBE-KJPAW pseudopotentials, a lattice constant of 4.004 Å, $4 \times 4 \times 1$ k -point mesh, and a Rydberg cutoff of 40 Ry as implemented in Quantum Espresso.³⁶ The bottom two layers of the slab were frozen in all calculations.

Foundational Machine Learned Interatomic Potential

For our foundation MLIP, we employ the MACE multi-head v0 model, that is trained on a broad mixture of materials, molecular, and catalytic configurations.^{21,23}

Pynta-driven Sampling

For both delta learning and finetuning it is vital to generate good and useful samples that are representative of the target chemical space to calculate as training data. This is traditionally especially challenging for transition states which are not trivial to generate and locate. Our workflow software Pynta for automatic calculation of kinetic parameters is able to automatically generate adsorbate and transition state guesses (along with optimizing and validating them). Using an MLIP, we run Pynta on our target space of reactions to generate a set of predicted and validated adsorbate and transition state geometries associated with the target space.

For our SIDT-driven delta learning we need DFT stationary point energies. Here we optimized each MLIP stationary point geometry at DFT level using ASE’s³⁷ QuasiNewton for adsorbates and Sella³⁸ for transition states and then calculated frequencies. Each stationary point was validated again at the DFT level based on their geometry and frequencies. In particular, we did not run DFT intrinsic reaction coordinate (IRC) calculations to confirm the DFT transition state connects to the correct reactants and products, which would have

dramatically increased computational expense. Instead, we assumed that if the MLIP IRC analysis was valid and the DFT geometry and frequencies aligned with the associated reaction then the transition state was valid. We successfully used a similar strategy to avoid excessive IRC calculations in our previous work on coverage dependence.^{16,33}

However, when finetuning MLIPs, training directly on stationary points is usually not optimal. As discussed in Schaaf et al., using MD-based sampling, rather than relying only on fully optimized geometries, yields a physically relevant distribution of local environments and makes training less sensitive to coherent geometric offsets introduced by the baseline model. Minimum geometries are for example sensitive to systematic shifts in bond distances that would appear identically across all minimized structures.²⁰ Thus, we generated our finetuning points by starting from the MLIP stationary point geometries and running short molecular dynamics (MD) simulations. All MD simulations ran 100 timesteps at 1 fs per timestep with a temperature of 600 K and a friction of 0.01 1/fs. For adsorbates that lie in attractive wells this was done normally. However, for transition states sampling the neighborhood of the stationary points is significantly harder, because in the absence of constraints structures will naturally drift away from saddles in the MD simulation giving samples far away from the target transition state.

While umbrella sampling and other metadynamics techniques are often used to overcome this problem, here we leveraged Pynta’s workflow, which intrinsically offers a solution to this problem. Pynta uses the harmonically forced saddle point search (HFSP) method to find transition state geometries.¹⁵ This method uses an SIDT to estimate the factor by which breaking and forming bonds will stretch in the transition state and defines harmonic potentials between those atoms that will cause an initial reactant configuration to optimize into a transition state guess. When Pynta searches for a saddle starting from that guess, the harmonic potentials are turned off and we optimize to a true saddle using Sella. To sample near transition state geometries we used Pynta’s initial guess for the successfully optimized transition states (determined by optimizing with the MLIP + harmonic potentials) and ran

MD simulations with the MLIP while keeping the harmonic potentials active, essentially artificially holding the structure in the vicinity of the saddle. This allowed us to easily draw near transition state samples for finetuning the MLIP.

Subgraph Isomorphic Decision Trees for Delta Learning

We represented our stationary points in RMG’s molecule framework as full adsorbate-slabs with a site-atom representation with breaking/forming bonds in transition states labeled as reaction bonds as done in our work on coverage dependence.^{16,33} Within this representation, in addition to RMG’s traditional site-atom representation, the site type, the site morphology, and the reaction information for transition states are made available.

We used a SIDT to learn the mapping between the stationary point graph structures and $H_{f0,DFT} - H_{f0,MLIP}$. While we tested a number of different SIDT decompositions, we achieved the best performance with a bond decomposition mirroring the structure of BACs. The used decomposition decomposes the prediction into a sum of contributions from each atom-atom and atom-site bond. The SIDT root structure was simply $R_x^*-R^*$, where R_x is any atom or site, R is any atom and the $*$ ’s label the target bond. We used two different kinds of initial root splits, the first was the same structure with the atom/site defined for the R_x and R for every unique pair of element and site (X-C, C-H, O-N, etc.), we will refer to this as the "element type splits." The second was split into 4 nodes based on whether the bond was a reaction bond and whether it was between a site and an atom or an atom and an atom, we will refer to this as the "bond type splits."

Results

Reaction Generation

We generated our chemical space on Pt(111) by starting from RMG’s primary Pt(111) thermochemistry library SurfaceThermoPt111 containing 153 surface species. We "reacted" all

adsorbates and pairs of adsorbates from that set using all of RMG’s surface reaction families. This gave us Pynta inputs for 8349 unique reactions. We also generated Pynta parameters for 126 unique surface diffusion reactions for those adsorbates yielding a total of 8475 possible reactions. We then sampled 20 test reactions from this set ensuring every family was represented and maximizing the 2-norm of the vector of the number of each unique breaking/forming bond type (e.g. N=C, O-H). We removed these 20 test reactions along with 4 ammonia decomposition reactions we already had calculations for on Pt(111)¹⁶ from the full set of reactions and calculated the 20 new test set reactions using Pynta with DFT. This left us with 8451 non-test reactions.

Foundational Model Pynta Run

We then calculated the 8451 non-test reactions using Pynta with the MACE foundation MLIP. Out of the 8451 proposed reactions, our workflow found at least one valid transition state for 608 reactions. The fraction of the successful reactions in each RMG reaction family is shown in Figure 1. Our analysis showed that around half of these failures occur because Pynta with the MACE foundation model does not find stable minima for all of the reactants and products for the reaction. This occurs for many physisorbed species for two tightly coupled reasons (1) the MLIP may not have a minimum near the proposed physisorbed geometry (2) Pynta’s own automated system for distinguishing between physisorbed and gas-phase species may disagree with the definitions used for species in the RMG library. Another significant portion of failures are reactions that make sense in RMG’s graph representation, but make significantly less sense in 3D. Additionally, as we will show later, another significant portion are not found because the foundational MLIP potential energy surface is not sufficiently accurate.

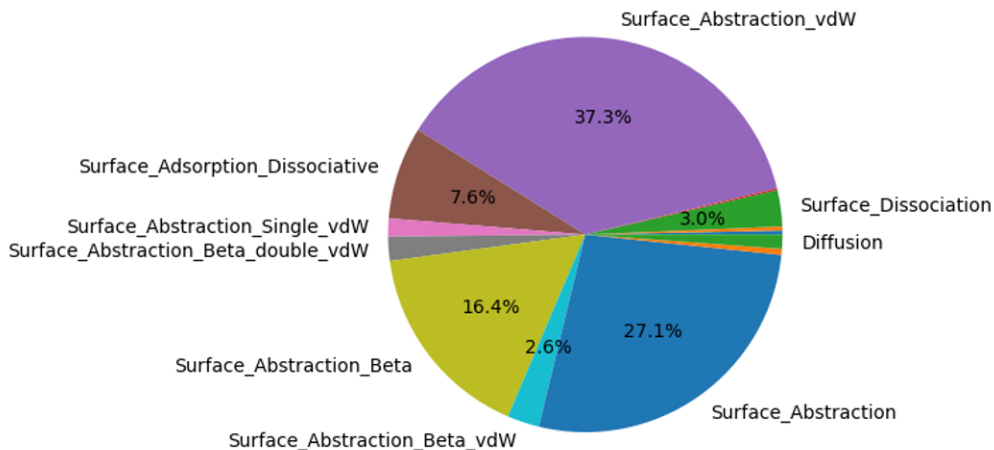


Figure 1: Pie chart of RMG reaction families^{1,2} in the set of successful reactions using the foundation MLIP.

Sampling

For finetuning the MLIP we calculated 12,800 near stationary point samples with 50% near TSs, 40% near adsorbate wells and 10% near gas phase wells, but otherwise drawn randomly from successful configurations from the foundation MLIP Pynta run. For training the delta-SIDT we optimized and calculated frequencies for all successful stationary points associated with the 608 successful reactions.

MLIP Finetuning

To improve the quantitative performance of the MLIP, we finetuned the MACE-MH0 foundation model with additional DFT data sampled as discussed earlier. Here we use the same DFT settings as the reference calculations, which are different to those used to train the foundation model. We adopt a straightforward continued-training procedure without replay,²¹ meaning that the pretrained weights are further optimized using only the newly generated dataset. Fine-tuning is performed with a learning rate of 10^{-3} and an exponential moving average of model parameters with decay 0.999.

Delta-SIDT Training

After running Pynta on the original successful reactions with the finetuned model, the delta model training datasets were generated by searching for isomorphic graph structures between the Foundation and Finetuned configurations and the DFT configurations. For the foundation model run (which was on all possible reactions) this gave 962 gas phase configurations, 184 adsorbate configurations and 758 transition state configurations. For the finetuned model run (on only the successful reactions from the foundation model run) this gave 61 gas phase configurations, 159 adsorbate configurations and 446 transition state configurations. To balance the training dataset we limited the foundation delta model gas phase training to the 61 gas phase configurations found by the finetuned model. This gave us 1003 datapoints for training the foundation delta model and 666 datapoints for training the finetuned delta model.

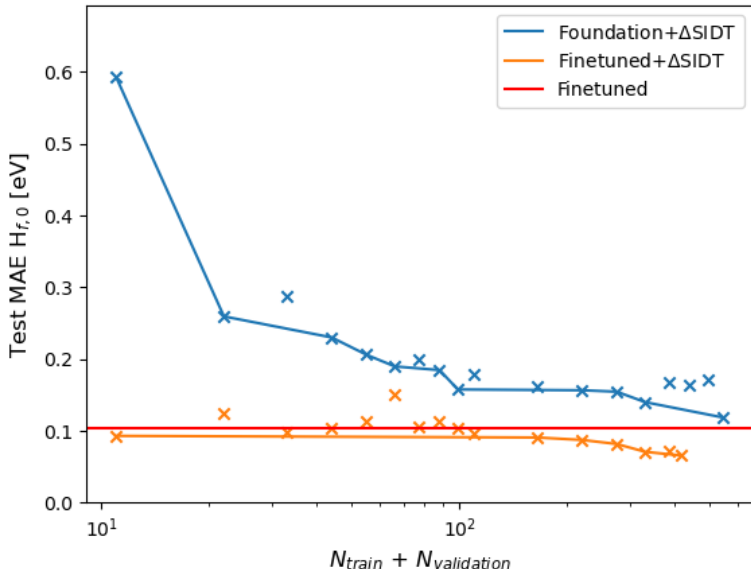


Figure 2: Learning curves for Δ SIDT models. The foundation model achieves 1.34 eV MAE on this test set putting it far above any point in this plot. The x-axis measures the number of species (which can correspond to more than one structure), not the number of structures although the two numbers are quite close for both the foundation and finetuning cases.

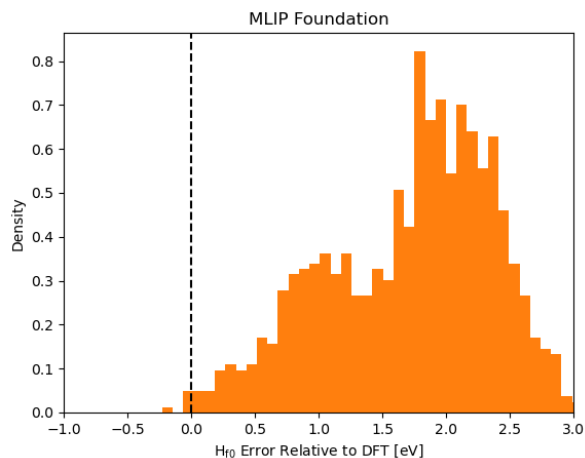
Figure 2 shows learning curves for training delta-SIDT models for both the foundation and finetuned MLIPs. All divisions were done based on species rather than structure to avoid

contamination. For both cases we set aside a 10% test set that is used to measure MAE for every point in Figure 2. The remainder was divided into randomly ordered 90% training and 10% validation sets. Key adsorbates H^* , O^* , HO^* , N^* , HN^* , C^* , and HC^* were forced to be included in the training set at the beginning of the training set. We then trained models on increasing fractions of the training/validation sets. For models with < 100 training points the training and validation sets were combined and trained out until the number of nodes was equal to the number of training points (not including the validation points added). For the other models, they were trained out until the number of nodes was equal to the number of training points and then regularized to a smaller model using the validation set. Then graphical regularization was applied and the test errors calculated.

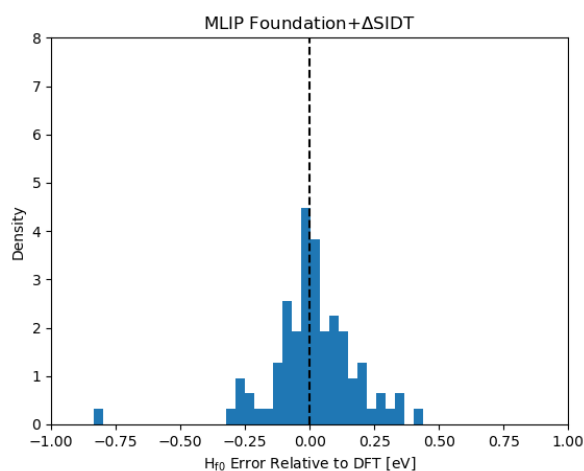
The MACE foundation model error on this test set is about 1.34 eV MAE and was thus not included in the plot. With the whole training/validation sets we see that the foundation delta model achieves similar performance to the finetuned model while the finetuned delta model is able to achieve significant performance improvements over the finetuned model reaching 0.07 eV MAE. With small training/validation sets the foundation delta model achieves enormous improvements over the MACE foundation model with diminishing returns after about 30 training points. While the initial performance improvements for the finetuned delta model over the finetuned model look small relative to the differences we see elsewhere in the plot it is a notably large improvement over the finetuned model that requires only a minimal number of training points. In practice for delta learning on very small datasets it is advisable to carefully select the calculated configurations to provide the best coverage of the chemical space, which should enable improvement over what is shown here.

Error Analysis

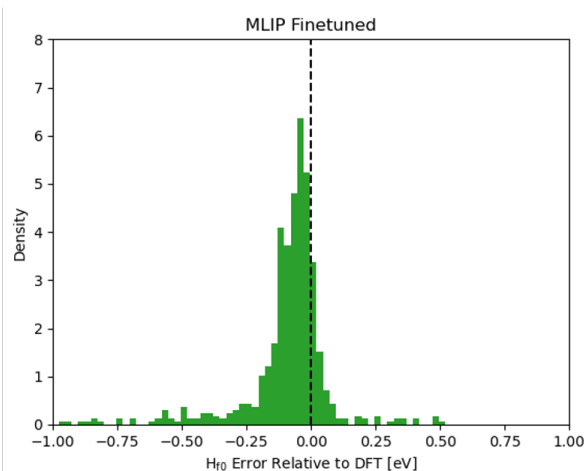
An error analysis of these frameworks on stationary point enthalpies is shown in Figure 3. The MLIP foundation model and MLIP finetuned model errors are with respect to all stationary point training data for that particular model while the MLIP foundation and



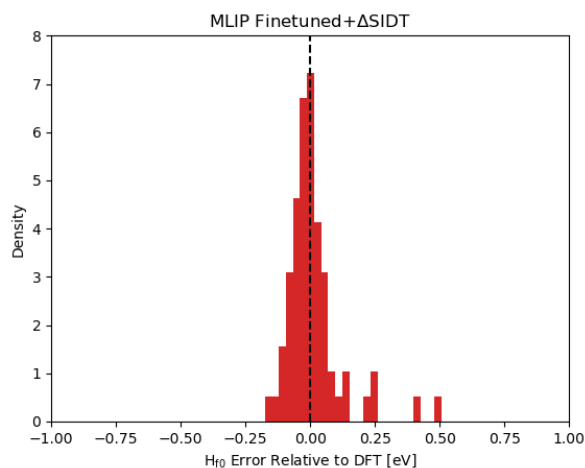
(a) Foundation model



(b) Foundation model with SIDT delta model



(c) Finetuned model



(d) Finetuned model with SIDT delta model

Figure 3: Error distributions on stationary point enthalpies for various approach combinations. Errors are all independent sets with respect to the SIDT delta models. For the foundation and finetuned models alone the errors are with respect to all stationary training points for that model. For approaches involving SIDT delta models the errors are with respect to the held out test set. The finetuned MLIP is not trained directly on the stationary points, but as the finetuning set is drawn by sampling near the stationary points being test

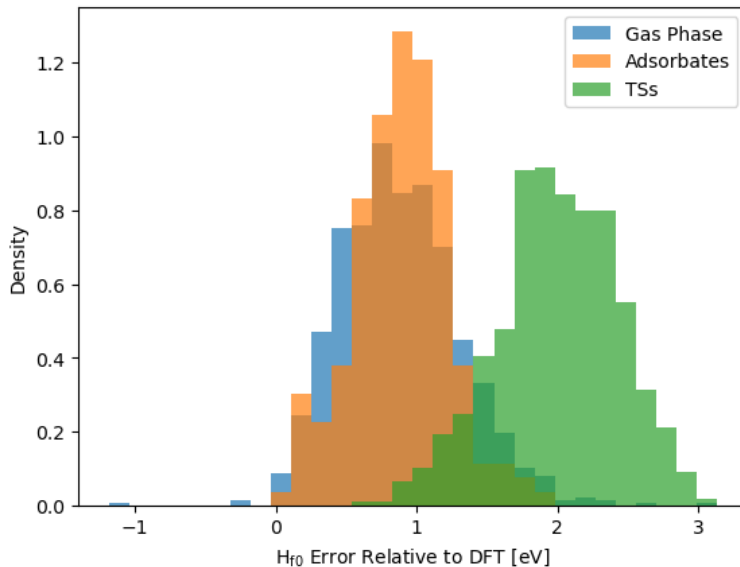


Figure 4: Error distributions on stationary point enthalpies for the foundation model split by stationary point type.

finetuned delta models are errors with respect to their held out test sets. It is worth reminding that the finetuned model’s training set includes points sampled near the stationary points used in this error analysis so the finetuned errors likely underestimate the true errors.

The foundation model errors have a strange shape that is best understood by looking at the distributions of different kinds of stationary points separately as shown in Figure 4. The foundation MLIP suffers from a large bias that differs between well/minima stationary points (gas phase and adsorbates) and transition states. For the minima this bias is about a 1 eV underestimation, while for the transition states it is about a 2 eV underestimation. It is worth noting that the stationary point biases are likely partially due to the MLIP being trained on a different DFT functional than BEEF-vdW, which we used for all calculations in this study. Nevertheless, the difference in the biases in the saddle points and the wells cannot simply be explained by differences in density functionals.

The foundation delta model and the finetuned model achieve similar MAE errors, but have very different distributions. While the foundation delta model is centered around zero error, it is appreciably wider than the finetuned distribution. At the same time the finetuned

distribution has an 0.09 eV bias toward overestimating enthalpies. This is interesting because the finetuned model was not observed to have any bias relative to the non-stationary point test configurations in finetuning and suggests that at least in this case the finetuned MLIP has a consistent bias when predicting enthalpies. The finetuned delta model gives the best errors being both better centered and thinner than the other models.

Comparison with Test Set Reactions

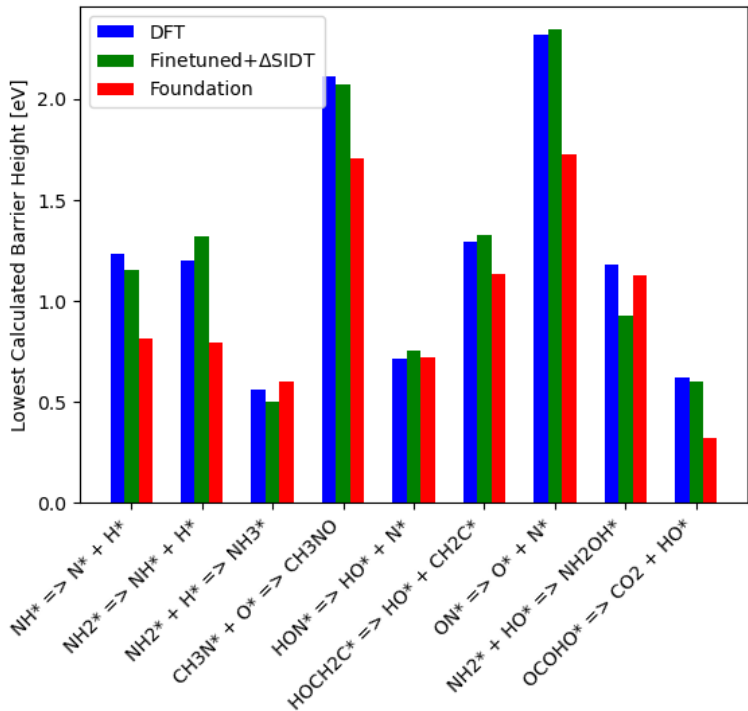


Figure 5: Comparison of barrier height predictions on test set reactions.

Since the finetuned model sampled configurations near all of the stationary points in the error analysis, it is important that we examine some fully held out configurations. Additionally, while the enthalpies of adsorbates are of interest here, there are far more transition states of interest in our chemical space and our primary quantity of interest is the associated barrier height.

Figure 5 shows a comparison of barrier height predictions on our fully held out test set reactions. It should be noted that while the transition states for these reactions were held out

the reactants/products were not. We set aside 24 test reactions, DFT calculations gave valid transition states for 14/24 of the reactions and the foundation and finetuned models both succeeded on 10/24 reactions, although only 9 reactions were found by both MLIPs. The MLIPs appeared in particular to have trouble with rare bond types and diffusion reactions. It should be noted that our test set was designed for diversity and is not distributed the same as the reactions in our chemical space so these success fractions should not be expected to be representative of the whole chemical space.

Overall we see that our combined finetuning and delta learning framework is able to achieve an overall 0.07 eV MAE relative to DFT on these barriers which is similar to its performance on the individual enthalpies. The foundation model, as expected, significantly underpredicts barrier heights overall. The combined finetuning and delta learning framework predicts all of the barriers within 0.12 eV except for the $NH_2^* + HO^* \rightarrow NH_2OH^*$ reaction where it underpredicts the barrier by 0.25 eV.

Finetuned Model Pynta Run

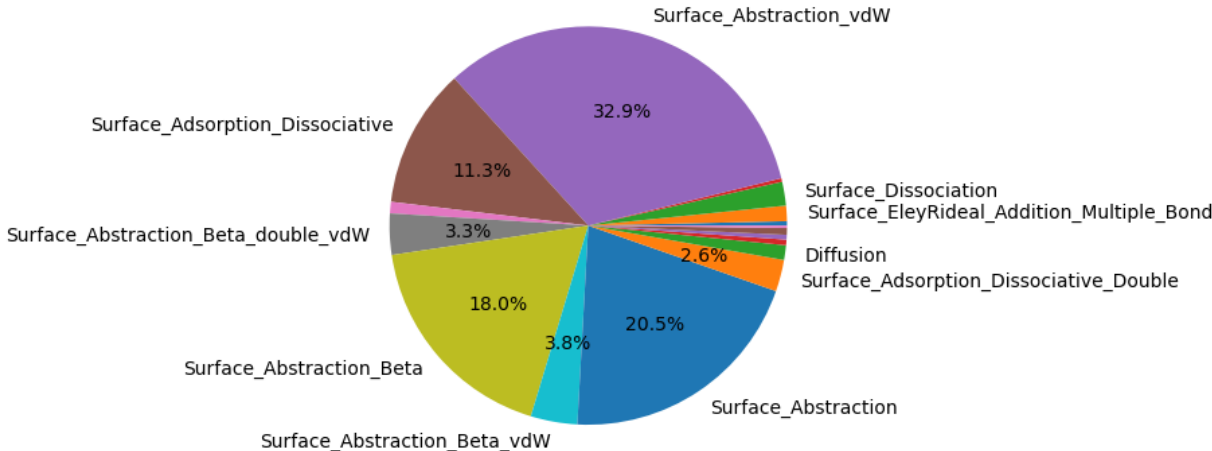


Figure 6: Pie chart of reaction families in the set of successful reactions using the finetuned MLIP.

Now that we have validated our finetuning and delta learning framework, we would

like to use it to calculate thermochemical and kinetic parameters for as many reactions in our chemical space as we can. We took our original chemical space of 8451 proposed reactions and attempted to calculate them all with the finetuned model. The finetuned model found transition states for 3111 of the proposed reactions, about 5x more than found with the original, foundational model. Giving us near DFT-accuracy rate coefficients for 3111 reactions on Pt(111).

It is important to note, however, that we do not know how many of these proposed reactions would have valid transition states at DFT. As with the foundation model, the finetuned model did not find reactants and products for about half of the reactions. If we were to trust performance on adsorbates that would imply the finetuned model found transition states for a majority of the remaining reactions. Of those remaining reactions there may be many for which that cannot occur. However, based on test set performance, one should suspect that there may be hundreds of reactions the finetuned did not find transition states for that could be found with DFT. Nevertheless, 3111 reactions is a very considerable portion of our target chemical space.

Discussion

Pynta-Driven Sampling

Using Pynta with a foundational MLIP on a large set of relevant reactions has proven to be a particularly practical way to generate finetuning and delta-learning samples. These Pynta runs provide foundation model predictions for gas phase and adsorbate wells and saddle points, which can be readily used as starting points for sampling. However, the most critical integrated innovation here (which was first done in Price et al.)²² is in the exploitation of Pynta’s HFSP parameters.

Transition states are traditionally challenging to sample near with MD because unlike at a well the MD simulation will tend to drift away from saddle points toward wells. However,

as mentioned earlier Pynta’s HFSP framework for generating saddle point guesses adds harmonic potentials to the potential energy surface to make the transition state guess the bottom of a well and optimizes to find the transition state guess. By MD sampling on the HFSP potential energy surface we are able to easily and effectively generate near transition state single points for finetuning.

Models and Dataset

Our combined finetuning and delta learning framework enables us to predict configurations and barrier heights for CHON species on Pt(111) within ~ 0.07 eV MAE. This is significantly smaller than typical DFT errors relative to experiments, making it a good surrogate for DFT. On our local cluster calculating rate coefficients for 10 reactions might take about a week. With this framework we are able to compute ~ 4000 reactions (where reactants/products exist) in a week representing a roughly 400x speed-up.

While we believe this framework is very accurate when it does find transition states, it is not robust for all reactions. There are a few particular cases of failed calculations that are worth mentioning. The first is reactions involving uncommon bonds breaking/forming. Dissociation reactions of diatomic molecules, which have been a challenge in similar prior work,²² seem to be a challenge for this framework as well: the finetuned model failed to find a transition state for N_2 dissociative adsorption reaction, which was one of the test reactions. Although, interestingly the MACE foundation model does not. Another class of reactions where calculations often fail are reactions with very low barriers like surface diffusions. These kinds of reactions can be challenging to handle even without the complications of machine learning frameworks, because low barriers make the search more sensitive to noise or other unwanted features of the potential, and even on well-behaving potentials the search is harder because of the increased possibility of the optimizer latching on to an undesired eigenmode of the local Hessian. In particular, while we would expect almost every monodentate chemisorbed adsorbate to have valid transition states for diffusion, this frame-

work finds only 37 diffusion reactions for the 72 monodentate chemisorbed adsorbates. If we remove the ammonia test set reactions (which were not randomly sampled), and ignore the difference in distributions between the test set and full chemical space we would (1) expect that DFT will fail on 1/2 proposed reactions and (2) estimate that the foundation model would succeed on 6/10 reactions DFT succeeds on and the finetuned model on 7/10 reactions DFT succeeds on. However, applying that analysis to the full chemical space we would expect the foundation model to find 2535 reactions not 608. If we were to take (1) only to be true that would suggest that only about 4225 proposed reactions are successful at DFT and would give the finetuned framework a $3111/4225 = 74\%$ success rate. However, without running DFT on a larger sample distributed identically to the full chemical space it is difficult to estimate this success rate accurately.

The dataset of 3111 reaction rate coefficients generated here is of high significance. Surface reaction rate coefficients are notoriously tedious and computationally expensive to calculate. For reference, RMG’s training databases for surface reactions collectively only currently contain 91 surface reactions. Once these rate coefficients are added to the RMG-database, our contribution will represent a factor of ~ 35 increase in the number of available training reactions for surface reaction classes in RMG!

Graph-Based Delta Learning

We have shown that lone SIDT-based delta learning frameworks from the foundation model are able to achieve accuracies on enthalpies similar to foundation model finetuning approaches with far fewer training points. In particular, we did not observe learning to asymptote at any training set size examined in this work. This is a significant result, but it is very important to consider the differences in computational cost for stationary point training data for the SIDT delta model and the non-stationary single points used to finetune the MLIP. While the cost of single point DFT calculations is relatively straightforward, the cost of stationary point calculations is much less so.

For our demonstrations, here we computed stationary point enthalpies by starting from the MLIP stationary point geometry, optimizing to the DFT stationary point, and calculating vibrational frequencies to get the zero-point-energy (ZPE). However, by sacrificing some accuracy in the enthalpies, the computational cost can be dramatically reduced. First, one can use the frequencies from the MLIP stationary point to approximate the ZPE, removing the need for DFT frequency calculations. Such ZPE errors may mostly cancel out in barrier height and enthalpy of reaction computations. A second even more dramatic improvement can be achieved by replacing the DFT optimization with a single-point DFT calculation at the MLIP stationary point. The accuracy of this approximation depends heavily on how similar the MLIP and DFT geometries are. However, it provides stationary point enthalpies at the cost of one single point calculation.

However, it is possible to reduce computational cost for these delta-models in ways that do not sacrifice accuracy. While our demonstration of delta learning (Figure 2) used mostly randomly drawn subsets, the cost / performance tradeoff should be able to be significantly improved by using curated sets of stationary points designed to maximize the bond diversity. For CHON adsorbates in adsorbate-site representations there are 30 bond types and including transition state reaction bonds there are a total of 44 bond types. So one should be able to cover all bond types in only 44 stationary point optimizations. These approaches should in the future be able to be used to improve the cost / performance trade-off over the demonstration provided here.

General Application of Workflow Strategy

In practice, one should first set aside commonly problematic and unique reactions like diatomic dissociations. While these can be accurately learned with these frameworks, they can require a significant amount of attention and/or compute,²² which can be disproportionate to the computational cost of simply computing them with DFT.

In this work we showed the power of both the finetuned and delta modeling strategies

alone before combining them. It is clear that each approach is powerful in its own way; therefore, it seems that it is most practical to approach this process iteratively in the future, finetuning the model first and then developing a delta model each iteration. There are a lot of benefits to calculating the delta-model training with a better finetuned model: optimizations require fewer steps and thus are cheaper and are more likely to graphically match the finetuned configurations making them valid delta-learning points.

Conclusions

In this work we have leveraged Pynta-driven sampling and combined MLIP foundation model finetuning and SIDT-based delta learning to enable rapid calculation of kinetic parameters for CHON reactions on Pt(111). We first demonstrated that SIDT-based delta models relative to foundation models alone can achieve comparable accuracies on enthalpies to foundation model finetuning approaches. Combining both MLIP foundation model finetuning and SIDT-based delta learning into one framework we able to predict barrier heights within 0.07 eV MAE of DFT calculations, significantly below typical DFT errors relative to experimental values. In particular, our framework enables calculation of reactions roughly 400x faster than using DFT. We leveraged this framework to calculate rate coefficients for 3111 surface reactions on Pt(111).

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